

Algorithmic Graph Theory QHM6704

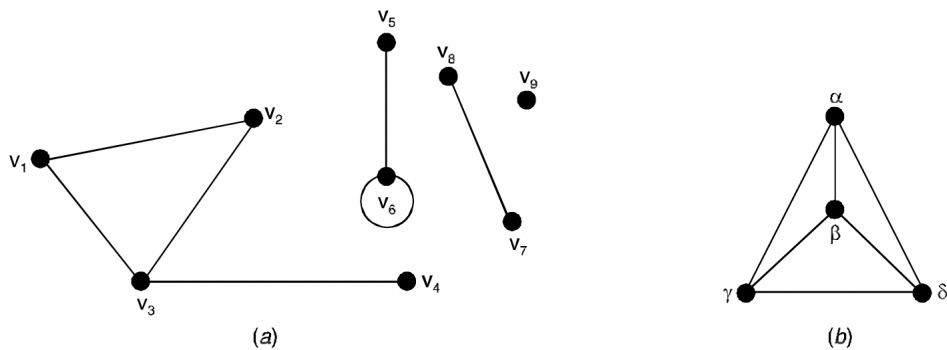
Problem Sheet

Dr Dimitris Chiotis

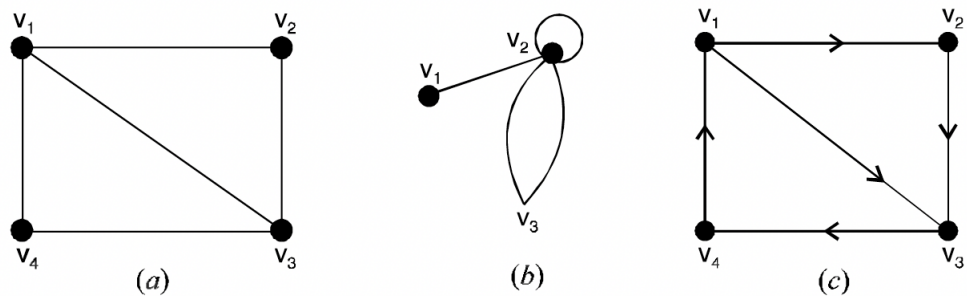
2025-2026

The following problems have been designed to help with comprehension of the key concepts. Please attempt them all. Most of them will be solved during the lab sessions and will be similar to what you will encounter at the mid-term test and the final exam.

Problem 1. Write down the vertex set and edge set of the graphs (a) and (b).



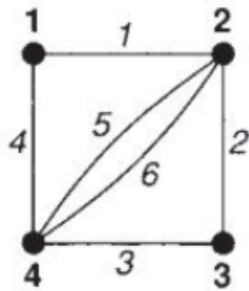
Problem 2. Find the adjacency matrix A for each of the graphs below



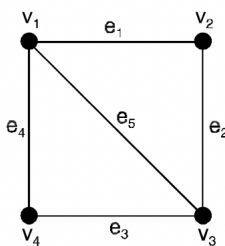
Problem 3. Draw the undirected graph with the adjacency matrix given by

$$A = \begin{pmatrix} 0 & 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 1 \end{pmatrix}$$

Problem 4. Find the adjacency matrix A and the incidence matrix M of the graph below



Problem 5. Find the incidence matrix of the graph below:



Degrees

Problem 6. Show that, in any gathering of six people, there are either three people who all know each other or three people none of whom knows either of the other two (six people at a party).

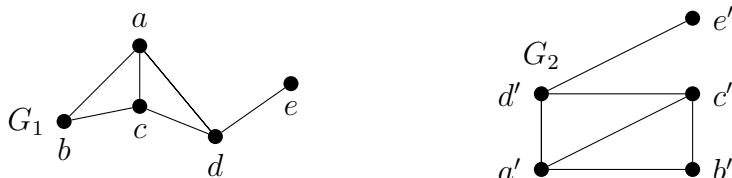
Problem 7. Determine the number of edges in a graph with 6 vertices, 2 of degree 4 and 4 of degree 2. Draw two such graphs.

Problem 8. How many vertices are needed to construct a graph with 6 edges in which each vertex is of degree 2?

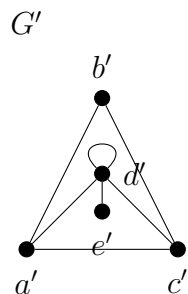
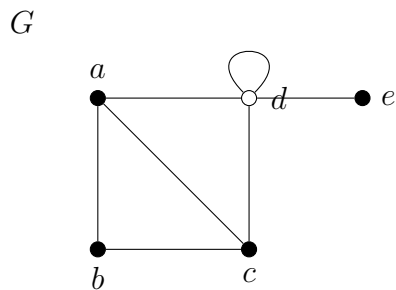
Problem 9. Is it possible to draw a simple graph with 4 vertices and 7 edges? Justify.

Graph isomorphism

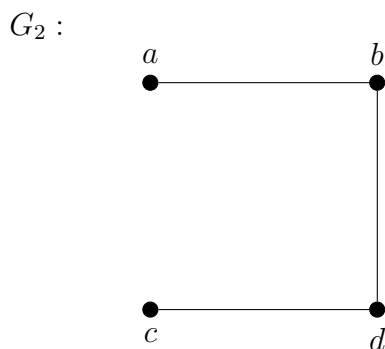
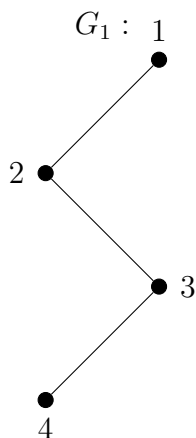
Problem 10. Show that the following graphs are isomorphic:



Problem 11. Show that the following graphs are isomorphic.



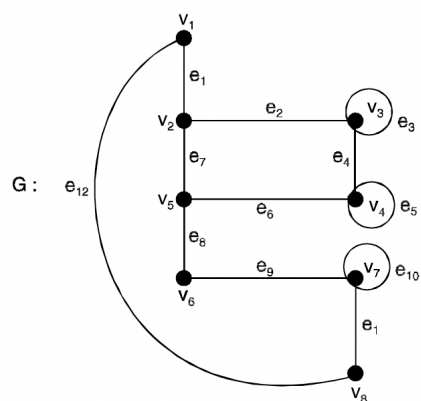
Problem 12. Show that the following graphs are isomorphic.



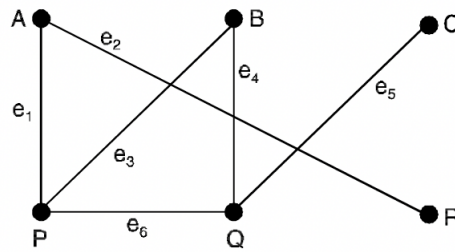
Walks, paths, cycles

Problem 13. Write down all possible

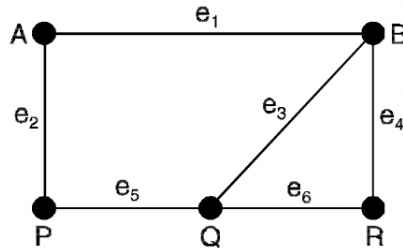
- (i) paths from v_1 to v_8 ;
- (ii) cycles of G ;
- (iii) trails of length three in G from v_3 to v_5 .



Problem 14. Find all cycles in the graph shown below:



Problem 15. Consider the graph shown below and find all paths from vertex A to vertex R . Also, indicate their lengths.



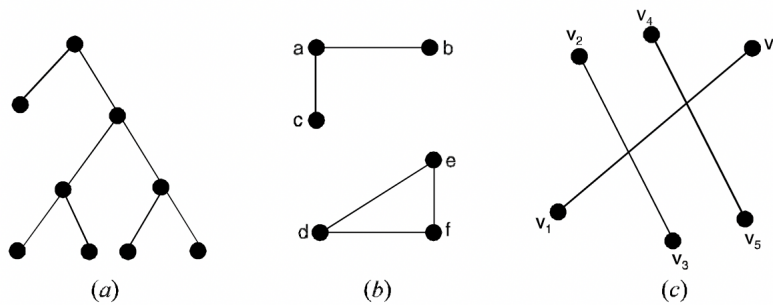
Problem 16. Prove the following:

- (i) A path with n vertices is of length $n - 1$.
- (ii) If a cycle has n vertices, it has n edges.
- (iii) The degree of every vertex in a cycle is two.

Problem 17. If G is a simple graph in which every vertex has degree at least k , prove that G contains a path of length at least k . Deduce that if $k \geq 2$, then G also contains a cycle of length at least $k + 1$.

Connectedness

Problem 18. Which of the following graphs are connected? Justify your answers.



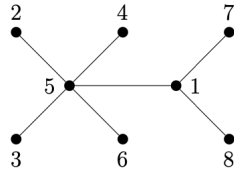
Problem 19. Show that if a and b are the only two odd degree vertices of a graph G , then a and b are connected in G .

Problem 20. Prove that a connected graph G remains connected after removing an edge e from G if and only if e lies in some cycle in G .

Problem 21. If G_1 and G_2 are (edge) decompositions of a connected graph G , then prove that $V(G_1) \cap V(G_2) \neq \emptyset$.

Trees

Problem 22. Determine the Prüfer code of the following tree



Problem 23. Draw the tree with Prüfer code $(3, 2, 3, 2, 3)$.

Problem 24. A tree has five vertices of degree 2, three vertices of degree 3, and four vertices of degree 4. How many vertices of degree 1 does it have?

Problem 25. A tree has $2n$ vertices of degree 1, $3n$ vertices of degree 2, and n vertices of degree 3. Determine the number of vertices and edges in the tree.

Algorithmic complexity

Problem 26. For functions $f, g, h : \mathbb{N} \rightarrow \mathbb{R}^+$, show the following:

- (a) (i) $f(n) + g(n)$ is $O(\max\{f(n), g(n)\})$.
- (ii) If $f(n)$ is $O(g(n))$ and $g(n)$ is $O(h(n))$, then $f(n)$ is $O(h(n))$.
- (b) Let $f(n) = a_0 + a_1n + a_2n^2 + a_3n^3$, where $a_0, a_1, a_2, a_3 \in \mathbb{Z}$ are constants. Show that $f(n)$ is $O(n^3)$.
- (c) Show that 2^{2n} is not $O(2^n)$. To this end, you may want to assume that it was and derive a contradiction.

Problem 27. Euclid's algorithm determines the greatest common divisor $\gcd(a, b)$ of two non-negative integers $a \geq b$ by setting $r_0 = a$, $r_1 = b$, and then repeating the following steps for rounds $n = 2, 3, 4, \dots$:

- If $r_{n-1} = 0$, then stop and output $\gcd(a, b) = r_{n-2}$.
- Otherwise, find q_n and r_n such that:

$$r_{n-2} = q_n r_{n-1} + r_n \quad \text{and} \quad 0 \leq r_n < r_{n-1}.$$

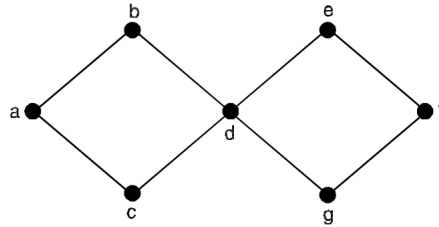
Note that $(r_n)_{n \geq 0}$ is a decreasing sequence of non-negative integers and the algorithm thus stops after a finite number of rounds. Note further that it makes sense to say that the size of the input of the algorithm is $\log_2 a + \log_2 b$, because this is the number of digits of a and b as binary numbers.

- (i) If r_n were to decrease by 1 in each round, how many rounds would the algorithm run for?
- (ii) Show that r_n in fact decreases significantly over two consecutive rounds of the algorithm, namely $r_n < r_{n-2}/2$. You may want to distinguish among the three cases where $r_{n-1} = r_{n-2}$, $r_{n-2}/2 < r_{n-1} < r_{n-2}$, or $r_{n-1} \leq r_{n-2}/2$.

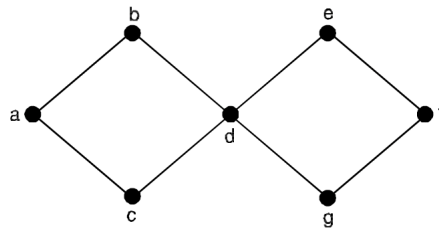
- (iii) Give an upper bound on the maximum number of rounds of the algorithm in terms of the size of the input to the problem. You may want to argue that after a certain number k of rounds, $r_k < 1$ and the algorithm must therefore have stopped.

Depth-first and Breadth-first search

Problem 28. Use the Breadth-first algorithm to find the spanning tree of the following graph

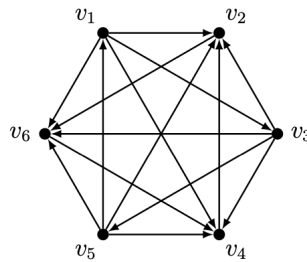


Problem 29. Find a spanning tree of the graph below using Depth-first search algorithm



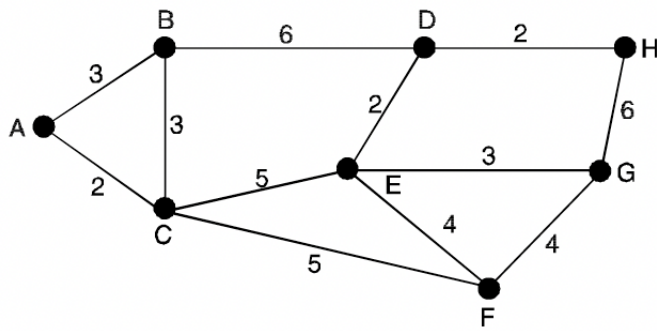
Strongly connected

Problem 30. Find the strongly connected components of the following digraph.

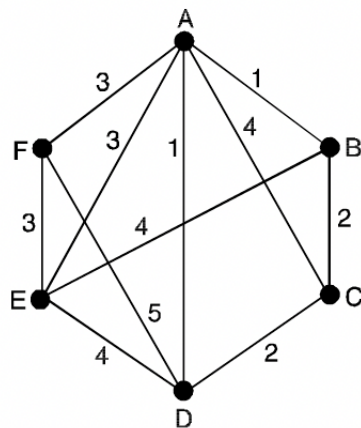


Minimum spanning trees

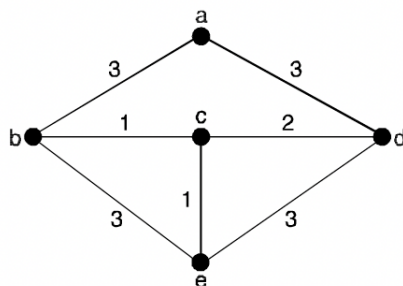
Problem 31. Use Prim's algorithm to find the minimum spanning tree of the following graph.



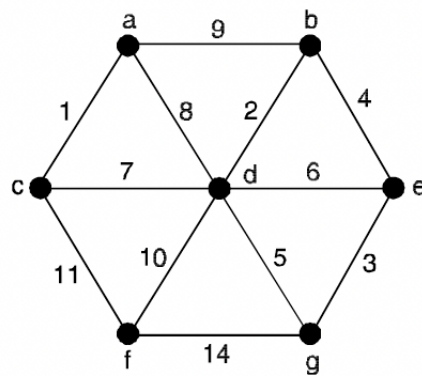
Problem 32. Use Prim's algorithm to find **two** minimum spanning trees of the graph below.



Problem 33. Use Kruskal's algorithm to find a minimum spanning tree of the following graph:

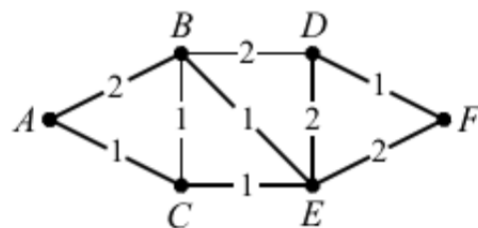


Problem 34. Use Kruskal's algorithm to determine a minimum spanning tree of the graph:

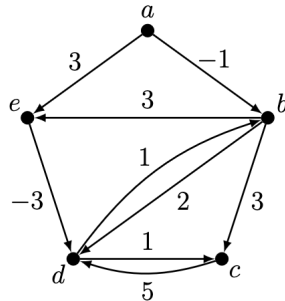


Shortest paths

Problem 35. Apply Dijkstra's algorithm to the following graph in order to find a minimum spanning tree with root A.



Problem 36. Utilising the Bellman-Ford algorithm, determine the shortest directed a-c-path in directed network given below.



Problem 37. Use the Bellman-Ford algorithm to find the shortest paths from vertex A :

